

PAPER_725: UQFF Quantum Variable Documents: S , $T_s^{\mu\nu}$, M_s , ω_s , B_s

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Class: UQFFKnowledgeBaseKB11 **CP4 Entry:** #309 **Keywords:** UQFF, spin angular momentum, stress-energy tensor, stellar mass, ω_s star spin, B_s magnetic field, U_{g3}
Session: 176 | **Version:** v5.33 **Source:** grok_share_ba508f76c8e.txt

Abstract

Five quantum variable documents define stellar and spin observables for UQFF: S (spin angular momentum of vacuum strings), $T_s^{\mu\nu}$ (stress-energy tensor for metric correction in the UP equation), M_s (source mass in U_{g2}), ω_s (stellar angular frequency for U_{g3} oscillation), and B_s (source magnetic field for string coupling). These connect UQFF to general relativistic metric corrections.

1. Variable Registry

Variable	Value	Physical Role
S	$5 \times 10^{-35} \text{ J}\cdot\text{s}$	Vacuum string spin: $g_{\mu\nu} + \eta T_s^{\mu\nu}$
$T_s^{\mu\nu}$	10^{-3} J/m^3	Stress-energy metric correction in UP(t)
M_s	$1.989 \times 10^{30} \text{ kg}$	Source mass in U_{g2} ($= 1M_{\odot}$)
ω_s	$2.5 \times 10^{-6} \text{ rad/s}$	U_{g3} oscillation: $\cos(\omega_s t \pi)$
B_s	10^{-4} T	Source field: $B_j(r, t) = \frac{\mu_0 \mu_j}{4\pi r^3} (1 + B_s \sin \omega_s t)$

2. Metric Correction

$$g_{UP} = g_{\mu\nu} + \eta \cdot T_s^{\mu\nu}(UA, SCm, SCm', RM, SM)$$

This metric term enters the Universal Permanence equation as the gravitational curvature contribution.

3. U_{g3} with Stellar Spin

$$U_{g3} = k_3 \sum_j B_j(r, \theta, t) \cdot \cos(\omega_s t \pi) \cdot P_{core} \cdot E_{react}$$

$$B_j(r, t) = \frac{\mu_0 \mu_j}{4\pi r^3} (1 + B_s \sin(\omega_s t))$$

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- UQFF KB grok_share_ba508f76c8e.txt entry #75
- Session 176, v5.33

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